Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 01)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

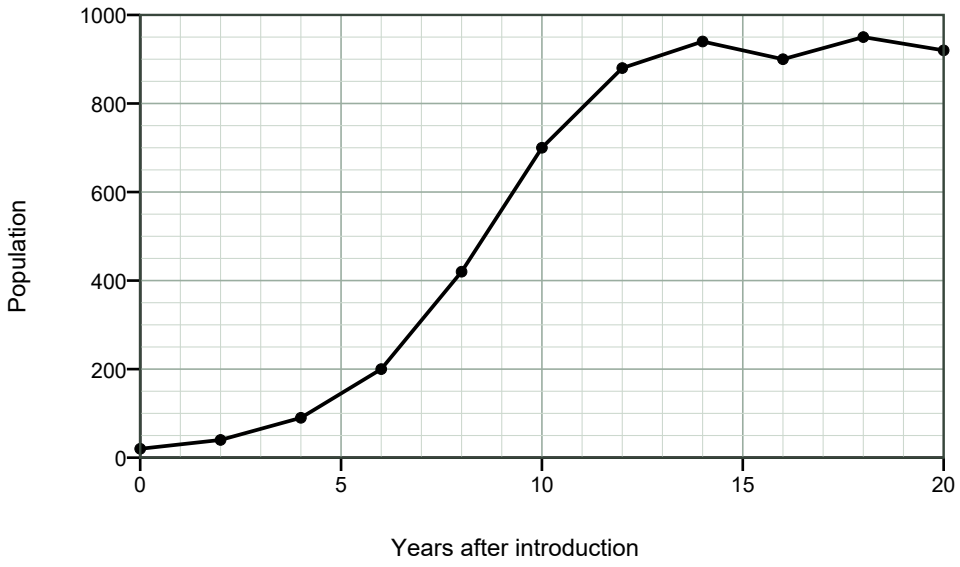
Question	Mark
1	
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TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 1 . 1 Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

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0 1 . 2 Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

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0 1 . 3 Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

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0 1 . 4 Explain how predator and prey populations regulate each other.

[2 marks]

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0 2 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 2 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 2 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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0 3 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 3 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 3 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 4 . 1 Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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0 4 . 2 Describe **five** features of a suitable release site.

[5 marks]

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0 4 . 3 Explain the importance of post-release monitoring.

[2 marks]

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0 5 . 1 Explain **three** features of natural systems that make them sustainable.

[3 marks]

0 5 . 2 Explain why the human use of mineral resources makes their sustainable exploitation much harder than the use of materials in a natural cycle.

[2 marks]

0 5 . 3 Explain **three** principles of the circular economy that should be applied when a new product is designed.

[3 marks]

0 5 . 4 Explain why an activity that is sustainable for a small population may be unsustainable for a large one. Use **one** example.

[3 marks]

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0 5 . 5 Explain why urban populations are less likely than rural communities to notice that a resource is being over-exploited, and give **two** consequences of this.

[4 marks]

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0 6 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 6 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

0 6 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 7 Table 1 shows some designated protected areas in the UK.

0 7 . 1 Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature	
Site of Special Scientific Interest (SSSI)			
Special Protection Area (SPA)			
Ramsar site			
Marine Conservation Zone (MCZ)			

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0 7 . 2 Explain how designating a protected area protects habitats and species.

[3 marks]

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0 7 . 3 Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

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0 8 Table 1 shows the energy inputs and food energy outputs of three farming systems, in GJ ha^{-1} yr^{-1}.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Subsistence maize, hand cultivated	0.6	9.0
Intensive wheat, UK	18.0	54.0
Intensive beef, feedlot	40.0	12.0

0 8 . 1 Calculate the energy ratio (output / input) for each system in Table 1.

[3 marks]

Answer

0 8 . 2 Explain what is meant by an **energy subsidy**, and use your answers to explain why the intensive systems have much lower energy ratios.

[3 marks]

0 8 . 3 Explain **three** ways a farmer can reduce the energy losses in a food production chain.

[3 marks]

0 8 . 4 Explain **two** advantages of genetic modification over conventional selective breeding.

[2 marks]

0 8 . 5 Explain **four** concerns about the use of GM crops.

[4 marks]

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0 9 The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 9 . 1 Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer ☐ thousand tonnes

0 9 . 2 Calculate the maximum catch that would have kept the stock constant.

[1 mark]

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Answer [ ] thousand tonnes

0 9 . 3 Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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0 9 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort .....

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Mean fish size and mean age .....

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1 0 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Extra space

1 0 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS